VICTORY 2027

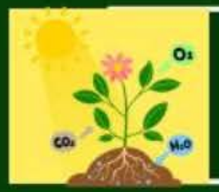
TRANSPIRATION

LECTURE-1 BIOLOGY

Upasana Ma'am



TOPICS TO BE COVERED



Concepts



Experiments



Uses




HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上



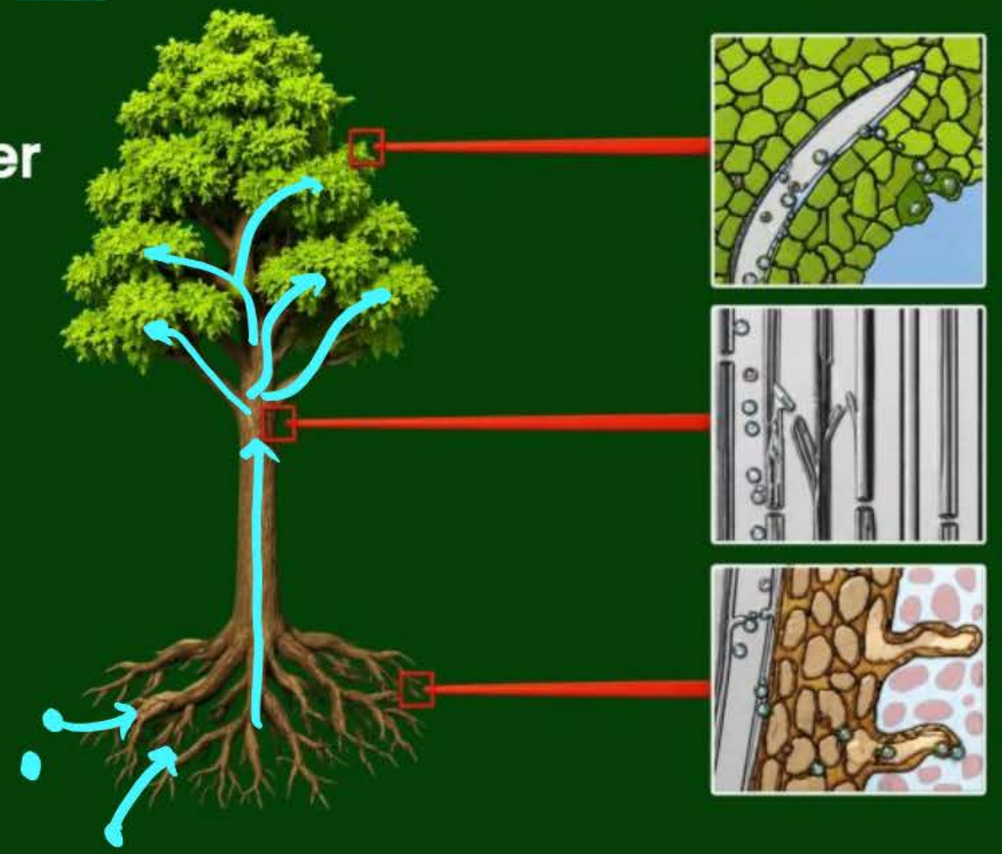
TRANSPIRATION

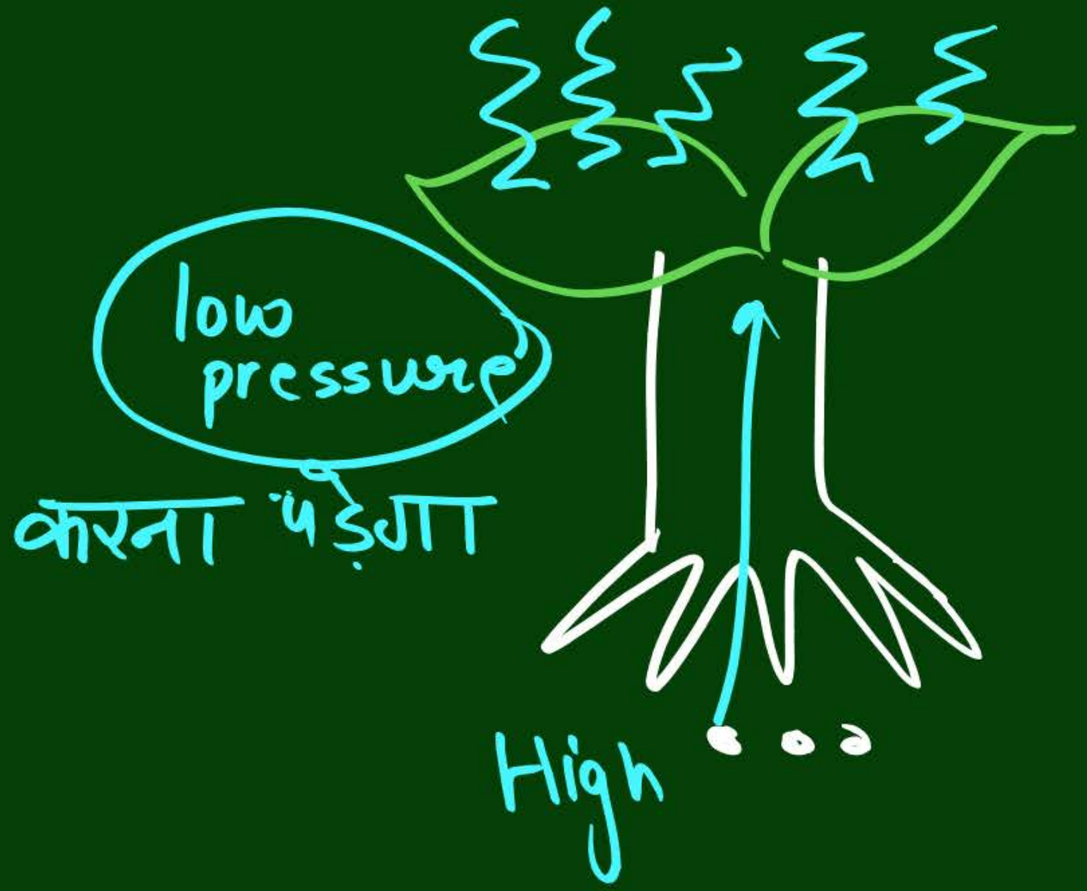
Transpiration is the process of loss of water in the form of water vapour from the leaves and other aerial parts of the plant.

Significance of Transpiration:

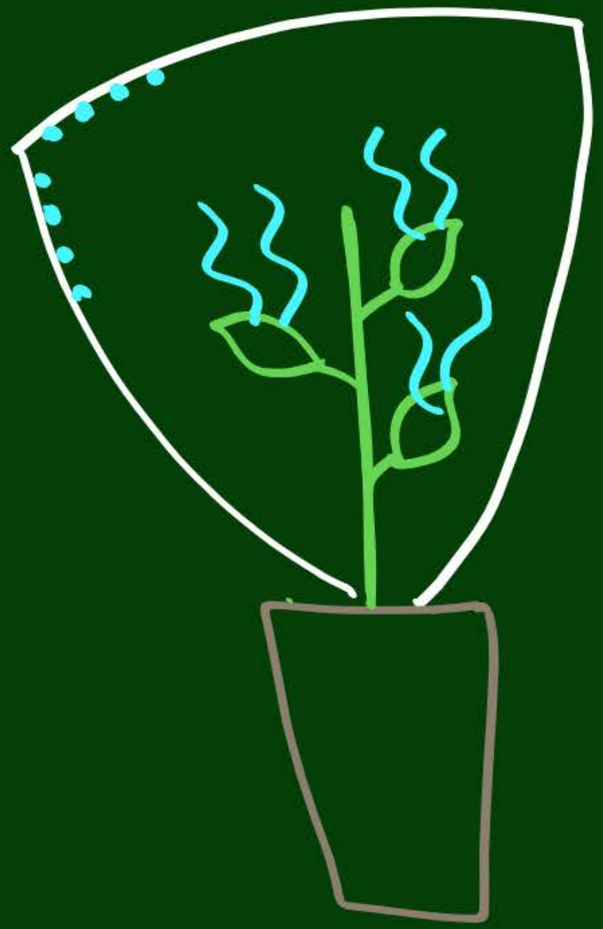
- Creating suction force to absorb water and minerals.
- For cooling effect in hot weather.

2% Needed




low pressure
करना पड़ेगा
High

DEMONSTRATION OF TRANSPIRATION





DEMONSTRATION OF TRANSPIRATION

No colour change
Control

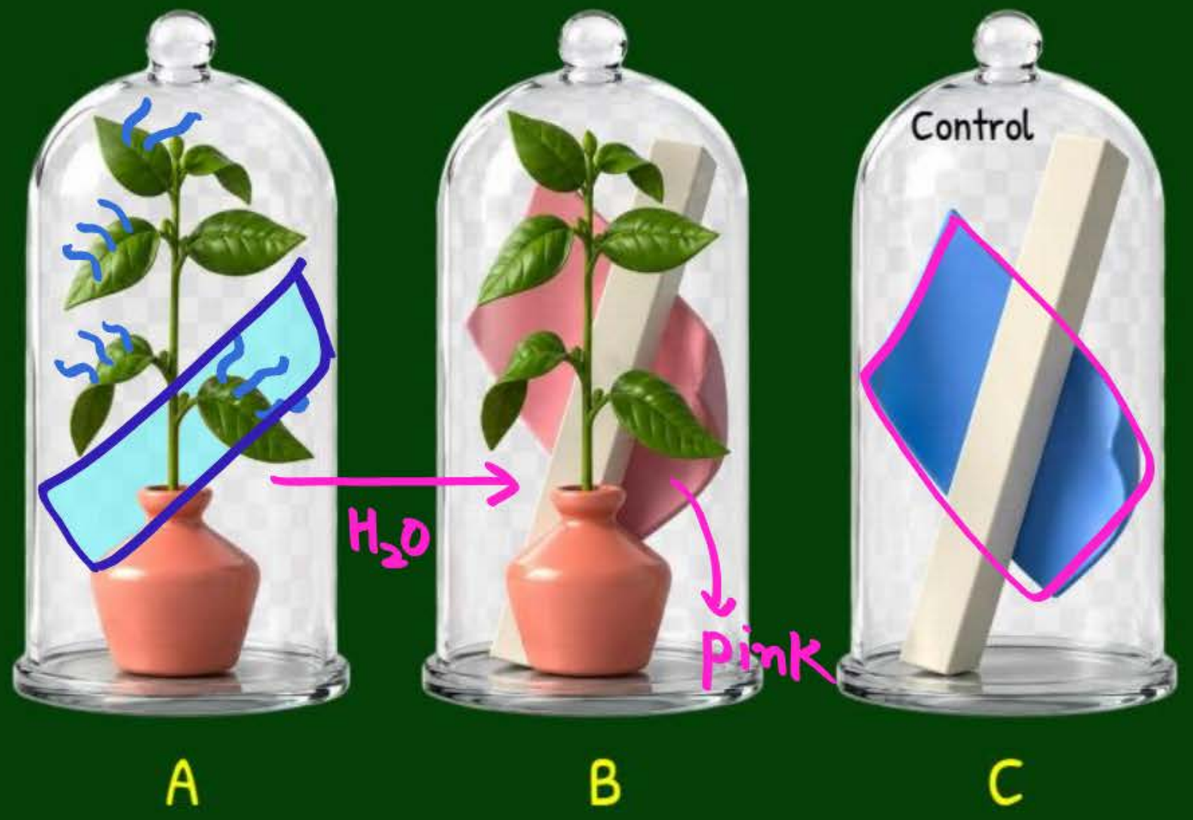
Cobalt Chloride Paper
Original Colour → Blue

↓

add H_2O

↓

Colour change to
Pink



The colour of cobalt chloride paper after contact with water?

A) Blue

B) Pink

Transpiration is highest during

A) Rainy Season

B) Winter

C) Summer

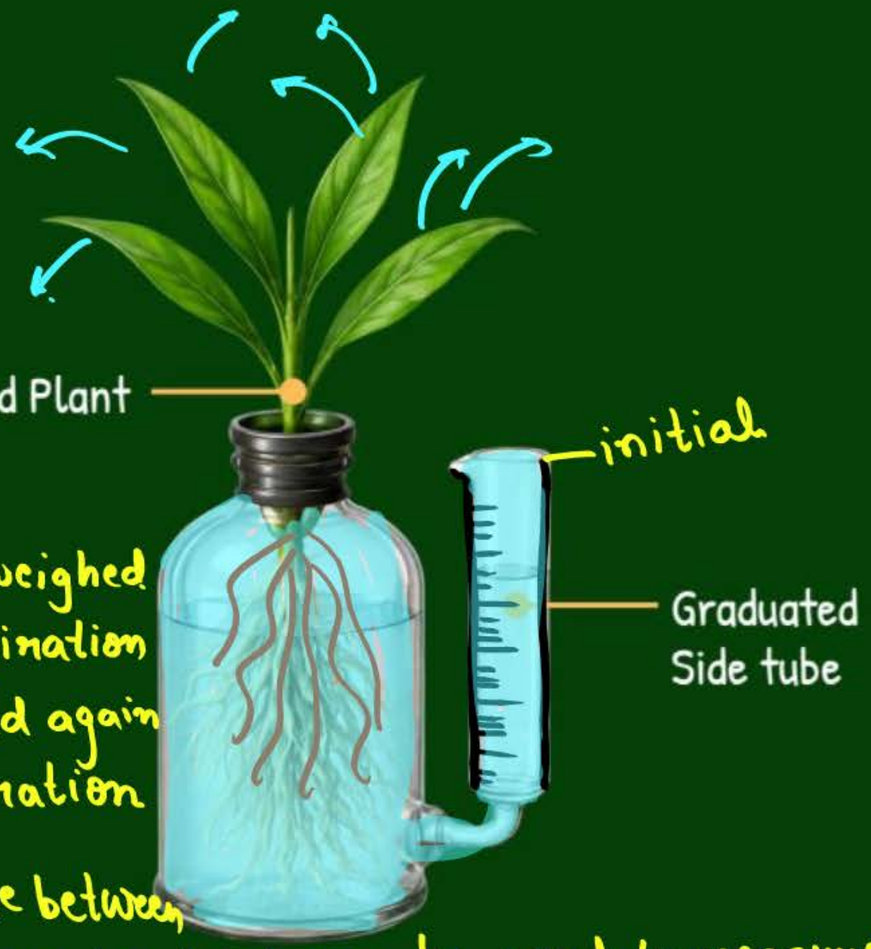
The permanent openings on stem is —

A) Stomata

B) lenticels

C) Cuticle.

MEASUREMENT OF TRANSPIRATION



The setup is weighed before transpiration & then weighed again after transpiration

The difference between the weight & water level can be used to measure transpiration



Weighing Machine

FLAMES

Me: *Water Plants Everyday*

Plants: *Lose 99% of the Water Via Transpiration*

Me:



Am I a joke to you?

1. WEIGHING METHOD

- A small light weight potted plant can be weighed before and after the end of a certain period of time.
- The soil surface and the pot should be fully covered to prevent evaporation from the surfaces other than the plant.
- The loss in weight by the plant during that time is due to the loss of water by transpiration.

2. POTOMETER METHOD

Potometer is a device that measures the rate of water intake by a plant (L. poton: drink, meter : measure), and this water intake is almost equal to the water lost through transpiration.

100% water uptake ≈ lost in transpiration 98ml

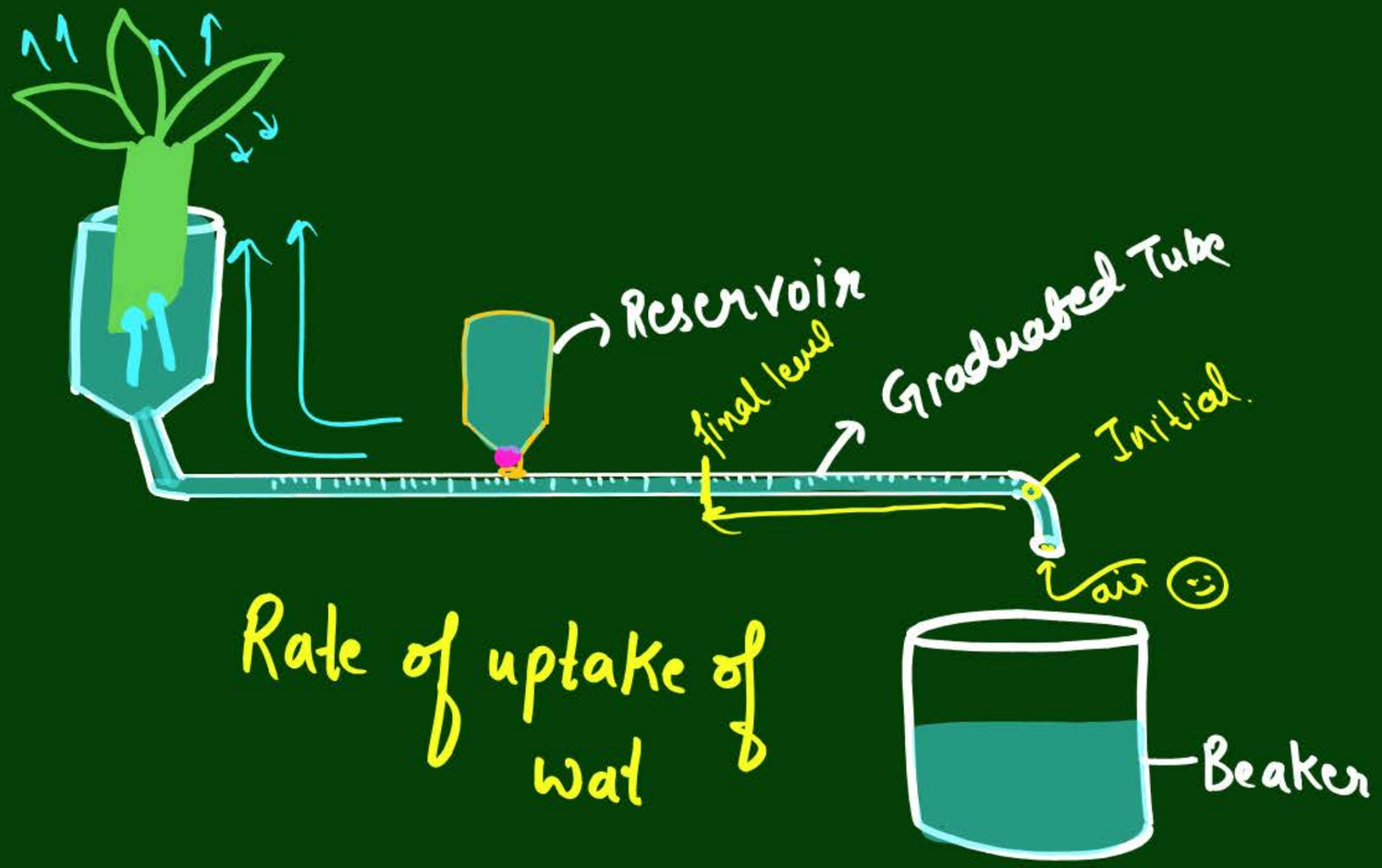
Extra.

Farmer's potometer – like Ganong's Potometer

Garreau's potomoter –

Measures transpiration from dorsiventral surface of leaf.

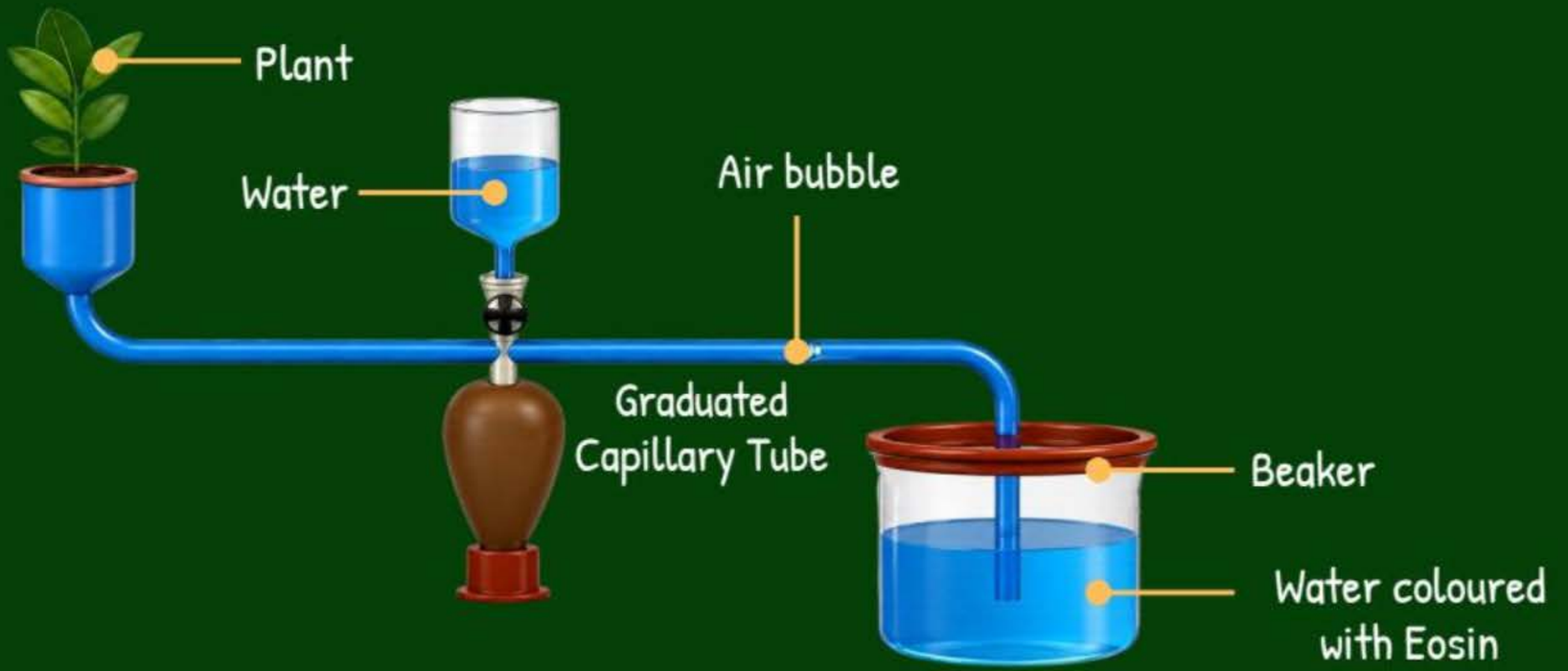




Rate of uptake of wat

3. GANONG'S POTOMETER METHOD

Potometer is a device that measures the rate of water intake by a plant (transpiration measure), and the water intake is almost equal to the water lost through transpiration.







PRECAUTIONS IN THE USE OF POTOMETER

- The potometer should be made completely water-tight.
- The twig should be cut obliquely (to allow larger surface for the water intake) and under water to avoid suction of an air bubble into the twig which will stop the absorption of water into the xylem.



LIMITATIONS IN THE USE OF POTOMETER

- Introducing the air bubble is not very easy.
- The twig may not remain fully alive for a long time.
- Any changes in the outside air temperature may affect the position of the air bubble in the capillary tube.



KINDS OF TRANSPIRATION



Stomatal transpiration

- Transpiration from the leaves through the stomata.



Cuticular transpiration

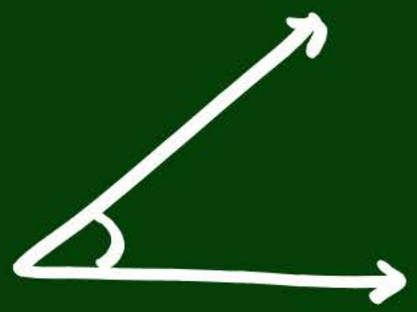
- Directly from the surface of the leaves and stems.



Lenticular transpiration

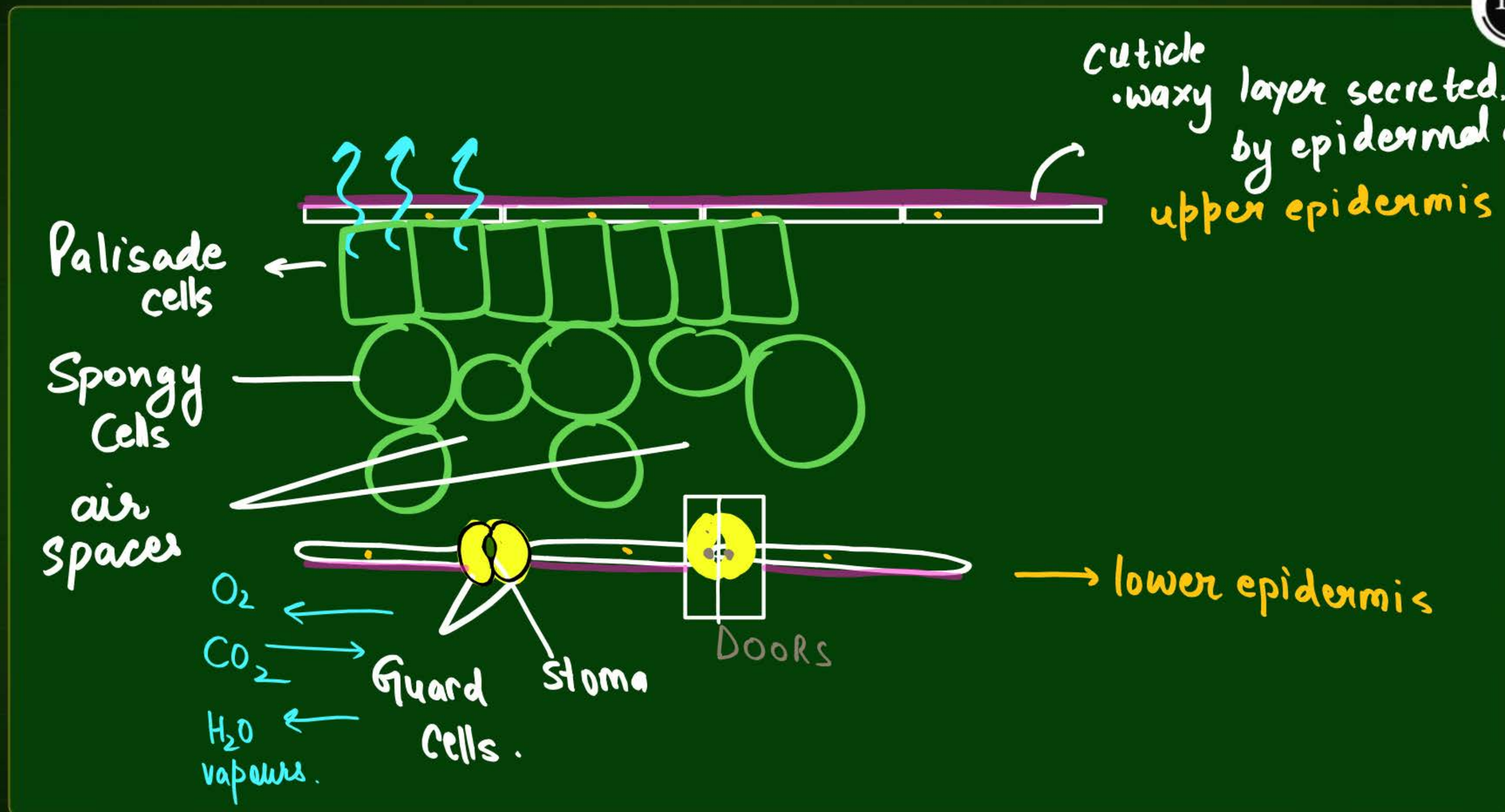
- From the lenticels which are the minute openings on the surface of old woody stems.

You guys are —



ie π

My NINJAS Cutie pie


Cuticle
.waxy layer secreted by epidermal cells
upper epidermis
Palisade cells
Spongy Cells
air spaces
lower epidermis
O_2
CO_2
H_2O vapours.
Guard Cells.
Stoma
DOORS

SUNDAY

one class

90% chances



Thank You